

Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be a locally Lipschitz function. Assume that f is convex.

We denote the subdifferential of f at x by

$$\partial f(x) = \{v: f(y) \geq f(x) + \langle v, y-x \rangle\}$$

(without convexity, we replace it by Clarke subdiff.)
We'll study asymptotic normality for stochastic subgrad. method

$$x_{k+1} = x_k - \alpha_k (V_k + \xi_k), \text{ where } V_k \in \partial f(x_k), \xi_k \text{ is noise sequence.}$$

Def (Active manifold):

Let $\bar{x}$ be a critical point of f ($0 \in \partial f(\bar{x})$).
 M is an active manifold of f at $\bar{x}$ if

1° (Smoothness)
Locally near $\bar{x}$, M is a smooth manifold containing $\bar{x}$, and $f|_M$ is a smooth function.

2° (Sharpness) $\exists$ neighborhood U of $\bar{x}$ and $\delta > 0$ s.t.

$$\text{dist}(0, \partial f(x)) \geq \delta, \forall x \in U \setminus M.$$

Examples:

1. $f(x) = \|x\|_2$. $M = \{0\}$

2. $f(x, y) = |x| + y^2$ $M = y\text{-axis}$
 $\{(0, y) : y \in \mathbb{R}\}$

3. $f(x) = \|x\|_2^2$. $M = \mathbb{R}^d$.

Lemma. Suppose that f admits an active manifold M at $\bar{x}$. then there exists $C > 0$ s.t. for any x near $\bar{x}$,

$$f(x) - f(p_M(x)) \geq C \|x - p_M(x)\|$$

Proof: (prop. 2.2 in the paper)

Lemma: Suppose that f admits an active manifold M at $\bar{x}$. Then, for any $y \in M$ near $\bar{x}$, $v \in \partial f(y)$

$$P_{T_M(y)}(v) = \nabla_M f(y).$$

Key assumption:

Strong- (α) regularity

We say that f is strongly (α) -regular along

M at $\bar{x}$ if there exists $C > 0$ s.t.

$$\| P_{T_M(y)} (\partial f(x) - \partial f(y)) \| \leq C \| x - y \|,$$

$\forall x$ near $\bar{x}$, $y \in M$ near $\bar{x}$.

$$\| P_{T_M(y)} (v - \overset{\nabla_M f(y)}{\hat{w}}) \| \leq C \| x - y \|,$$

$\forall x$ near $\bar{x}$, $y \in M$ near $\bar{x}$, and

$\forall v \in \partial f(x), w \in \partial f(y)$

Proposition (Davis - Drusvyatskiy - J., saddle paper)

① For any "definable" function $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$,

for all most every $v \in \mathbb{R}^d$,

$f_v := f(x) - \langle v, x \rangle$ only has finitely many critical points, admits active manifold at each critical point with strong (a) holds

② the nonlinear programming admits an active manifold with strong (a) (with assumptions in slides)

Main assumptions.

1. $\bar{x}$ is a critical point of f and f admits an active manifold at $\bar{x}$ satisfying strong (a)
2. Locally, f has quadratic growth at $\bar{x}$, i.e., for x near $\bar{x}$
$$f(x) - f(\bar{x}) \geq c \|x - \bar{x}\|^2.$$
3. SGD sequence $x_n \xrightarrow{\text{a.s.}} \bar{x}$.

Key ingredients of the proof.

$$x_{k+1} = x_k - \alpha_k (V_k + \xi_k), \quad \alpha_k = \frac{C}{k^r}, \quad r \in (\frac{1}{2}, 1).$$

① Faster speed towards manifold.

Note that the typical rate of $\|x_k - \bar{x}\|$ is $\alpha_k^{\frac{1}{2}}$.

$$\begin{aligned} \text{dist}^2(x_{k+1}, M) &\leq \|x_{k+1} - p_M(x_k)\|^2 \\ &= \|x_k - p_M(x_k) - \alpha_k (V_k + \xi_k)\|^2 \\ &= \|x_k - p_M(x_k)\|^2 - 2\alpha_k \langle V_k + \xi_k, x_k - p_M(x_k) \rangle \\ &\quad + \alpha_k^2 \|V_k + \xi_k\|^2 \end{aligned}$$

$$\mathbb{E} [\text{dist}^2(x_{k+1}, M) | \mathcal{F}_k]$$

$$\begin{aligned} &\leq \text{dist}^2(x_k, M) - 2\alpha_k \underbrace{\langle V_k, x_k - p_M(x_k) \rangle}_{\geq f(x_k) - f(p_M(x_k))} \\ &\quad + C\alpha_k^2 \\ &\geq \mu \|x_k - p_M(x_k)\| \\ &= \mu \text{dist}(x_k, M) \end{aligned}$$

$$\leq \text{dist}^2(x_k, M) - 2\mu\alpha_k \text{dist}(x_k, M)$$

$$x^2 = x^2 - \mu\alpha_k(x) + C\alpha_k^2$$

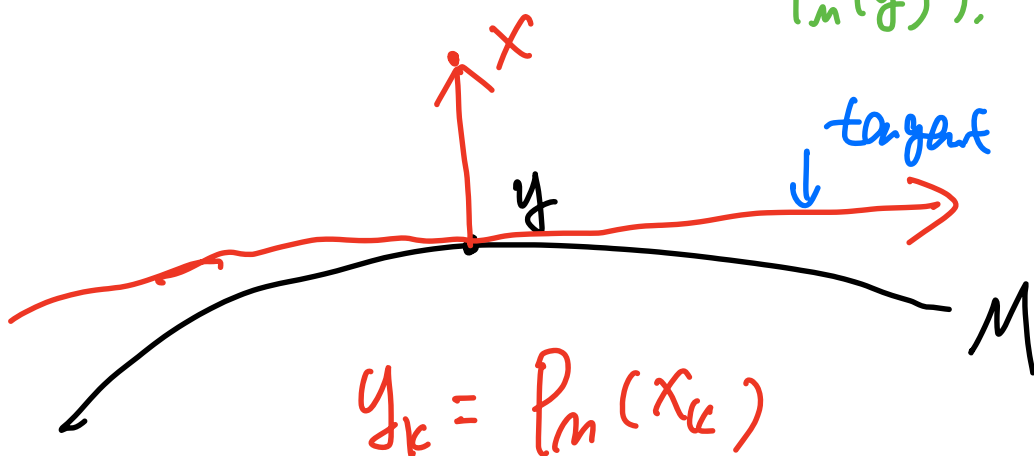
We expect $\text{dist}(x_k, M) \lesssim \alpha_k$. $\ll \alpha_k^{\frac{1}{2}}$

② Dynamics on the manifold

let $y_k = P_M(x_k)$.

Fact: P_M is smooth. $\forall y \in M$,

$\nabla P_M(y) = P_{T_M(y)}$ (orthogonal proj. onto $T_M(y)$).



$$y_{k+1} = P_M(x_k - \alpha_k(V_k + \xi_k))$$

$$= P_M(y_k + \underbrace{(x_k - y_k)}_1 - \alpha_k(V_k + \xi_k))$$

$$= P_M(y_k) + \underbrace{P_{T_M(y_k)}}_{\nabla P_M(y_k)} \left((x_k - y_k) - \alpha_k (v_k + \xi_k) \right) + O(\|x_k - y_k\|^2 + \alpha_k^2)$$

$$= P_M(y_k) - \alpha_k P_{T_M(y_k)}(v_k) - \alpha_k P_{T_M(y_k)}(\xi_k) + O(\alpha_k^2)$$

Strong (a)

$$= P_M(y_k) - \alpha_k \nabla_M f(y_k) - \alpha_k \|x_k - y_k\| - \alpha_k P_{T_M(y_k)}(\xi_k) + O(\alpha_k^2)$$

$$= \underbrace{P_M(y_k) - \alpha_k \nabla_M f(y_k) - \alpha_k P_{T_M(y_k)}(\xi_k)}_{\text{Riemannian SGD}} + O(\alpha_k^2)$$

Riemannian SGD